110 MINUTES OR TWO 55 MINUTE PERIODS

Make a fitted tray

A complete first-day job using personal tape measures, pocket notebooks, and ordinary paper supplies.

Tomorrow's result: each student designs a tray for their own notebook, builds a partner's tray from a work order, checks the fit, and records a revision. Use it as a low-stakes baseline for measurement, math, and communication.

Ten minutes of preparation

Set out about **3 sheets of letter paper per student**, scissors and a straightedge per pair, shared tape, and pencils. Print one double-sided student job sheet per pair, or display it. Make one sample using the example on page 2. Students use their own tapes and notebooks. A folder edge can serve as a straightedge.

Set the job: a shallow tray that sits on a desk and allows its owner to put the notebook in and remove it. Start with **1/8 inch clearance at each side** and **3/4-inch walls**. This is a thin-paper fit prototype; it is not a carrying tray. Do not tape or cut students' notebooks.

Minutes	Students do	You check or teach
0-7	Handle the sample; name what must fit.	State the work order and show the acceptance test.
7-15	Each marks a 2 3/8 in line on paper.	Watch zero, alignment, tick reading. Use the quick check on p. 3.
15-27	Measure closed notebook length and width. Partner rechecks.	Record actual units and any disagreement; remeasure before choosing a value.
27-40	Calculate base and blank dimensions.	Model two clearances and two walls. Use p. 2.
40-50	Write a work order and small labelled sketch.	Approve arithmetic and sheet fit before construction.
50-55	Swap work orders; clarify missing information.	Show the four cuts and tab folds. Pause here if there is a bell.
55-78	Draw the full-size pattern; cut, fold, and tape the partner's tray.	Each student builds. Demonstrate the first corner slowly.
78-88	Return trays to owners; inspect and test.	Record an actual result, including mismatches.
88-99	Correct a fit/fold/label problem and retest.	Require a visible change or a justified decision to retain the dimensions.
99-106	Complete the individual exit on student p. 2.	Keep this independent; collect/read a small evidence set.
106-110	Clean up and label work.	Keep the notebook record and final tray or flattened pattern.

Opening words: “Your math has a job today: it has to make a part fit. Someone else must be able to build from your notes. We will use these notebooks to record our work all year.”

Teach the allowance

Keep the two side clearances separate from the two folded walls.

Teacher example only	Calculation	Result
Notebook	Use a model notebook 5 1/2 in long and 3 1/2 in wide.	Students use their own measurements.
Inside base length	5 1/2 + 1/8 + 1/8	5 3/4 in
Inside base width	3 1/2 + 1/8 + 1/8	3 3/4 in
Blank length	5 3/4 + 3/4 + 3/4	7 1/4 in
Blank width	3 3/4 + 3/4 + 3/4	5 1/4 in

Say: “A gap is needed at both opposite edges, so 1/8 + 1/8 = 1/4 inch. We also need a wall at each end: 3/4 + 3/4 = 1 1/2 inches. Those walls are part of the flat sheet.”

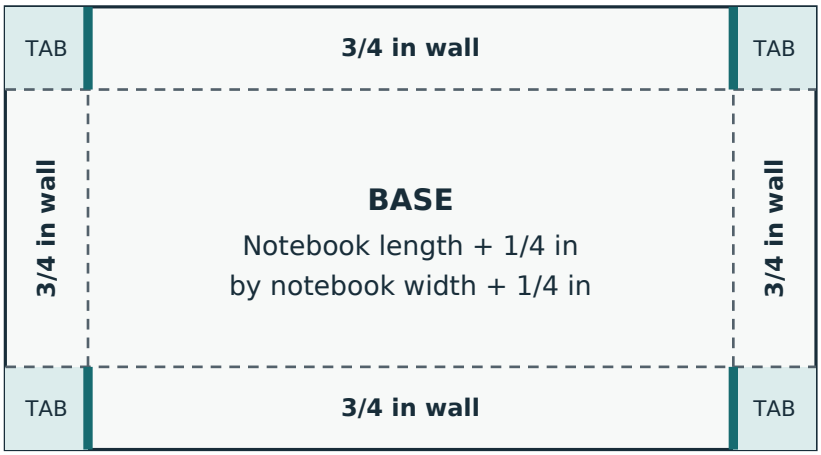

Diagram is NTS. Draw your own full-size pattern from your calculated dimensions.

Demonstrate before they cut

1. Draw and cut out the calculated outer rectangle on loose paper. Measure 3/4 inch inward from all four edges; draw the fold lines with a straightedge.
2. Make only the four short cuts shown in the diagram. Stop at the first fold line. Keep the corner tabs.
3. Pre-fold the four walls and tabs. Lift the walls; turn each tab inward against its neighboring wall. Tape the corners while holding them square.
4. Set the tray flat and test it with the owner's notebook. Do not force it into place.

Before building: the work order must identify the owner, base length and width, blank length and width, wall height, units, and a labelled sketch. Compare the blank with the actual sheet in both orientations. If it will not fit, join two sheets and record the seam, or agree on a smaller test object and recalculate. Do not silently change the wall height.

What to look for

A successful first day gives every student a finished attempt and gives you a usable baseline.

Readiness and exit keys

Opening: 2 3/8 inches is 2 inches plus six 1/16-inch spaces, or three 1/8-inch spaces. Read the actual tape divisions. For students who need another check, ask: a piece starts at 1 inch and ends at 4 5/8 inches; its length is **3 5/8 inches**. Do not teach thirty-seconds unless the tool and task need them.

Exit 1: notebook width 3 7/8 in plus two 1/8-in gaps gives base width **4 1/8 in**. Add two 3/4-in walls: blank width **5 5/8 in**.
Exit 2: adding one wall instead of two makes the blank **3/4 in too narrow**. If folded as planned, it leaves too little base width.
Exit 3: accept a specific observed result linked to a dimension, fold, corner, or drawing correction. If no size correction was needed, require the completed test result and one clarified label/instruction.

Acceptance check

With the tray fully supported on a desk, the owner inserts and removes the notebook **three times** without forcing it or tearing the corners. Check that the notebook lies flat, that wall folds follow the work order, and that the builder can point to the matching dimensions. The 1/8-inch gaps are starting design allowances; do not claim flexible paper demonstrates precision machining tolerances.

Capture four small pieces of individual evidence

Evidence	Independent	Needs a cue or more evidence
Tape use	Starts at the intended point; reads the appropriate fraction with units.	Confuses an endpoint reading with length, or counts the wrong divisions.
Mathematics	Adds both clearances and both walls; maintains units.	Adds an allowance once, or needs fraction support.
Handoff	Another student identifies the dimensions and builds from the work order.	Missing unit, base/blank distinction, or ambiguous sketch.
Check and revision	Reports the actual fit and links a change to an observed issue.	Reports only “good/bad” or changes the drawing without explaining why.

Record **independent / with cue / not yet observed** beside each student's name on your roster. Tomorrow is a baseline; missing evidence is not a fabricated low score. Do not let a partner's calculation stand in for the individual exit.

Keep everyone working

Support: use a fraction strip and a blank net with no numbers. Demonstrate on a separate scrap. Let the student dictate labels or use an adapted cutting partner while you observe their own measuring and math.
Extend: can two example blanks, each 7 1/4 by 5 1/4 inches, fit on one 8 1/2 by 11 inch sheet? **Yes:** put the 7 1/4 side across the 8 1/2 side and two 5 1/4 lengths along the 11 side: 10 1/2 inches. Draw the arrangement before cutting.
If time slips: simplify decoration and the sketch, preserve one build, one fit test, and the individual exit. A failed prototype with a clear correction is useful evidence.